

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

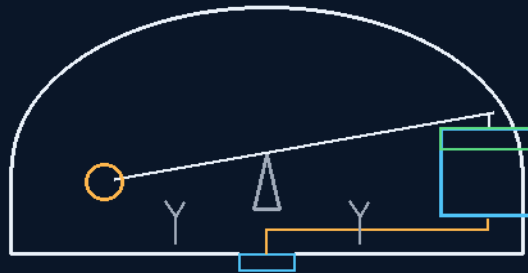
INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 293

THE SNEAKY THERMAL ENGINE CAPILLARY-WICK EVAPORATION ENGINE

closed room, breathing gradient
the bimetal band is the fuel line



perpetual operation — never perpetual creation

2-IN WICK · 10-TON RELEASE · 500 KJ A CYCLE

THE TAP HAS A NAME

Power demonstrator — v1.0 in force

GP-IM-293

Revision v1.0 — 23 August 2026

188 of 290

GRADED IN SITTINGS 581-582 — AMBIENT-THERMAL ENGINE

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LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — INSTRUCTIONS MANUAL — PHASE 293

PHASE 293: CLOSED-ROOM CAPILLARY-WICK EVAPORATION ENGINE AND BIMETAL-BAND UNLOADER — 'THE SNEAKY THERMAL ENGINE' — STATUS: CURRENT — GRADED IN SITTINGS 581-582: AN

AMBIENT-THERMAL ENGINE — PERPETUAL OPERATION, NEVER PERPETUAL CREATION. A power demonstrator, earth-side as much as ship-side: a sealed, moisture-locked clear-polycarbonate room; a capillary wick lifts the sump water; a bimetal band unloads it into a hanging IBC; when the tank's weight trips ten tons the lever falls five metres and spins the flywheel. The tap has a name — the room's breathing thermal gradient. In a truly isothermal room it stalls: affirmed by the audit, conceded by the Author.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Manual GP-IM-293, v1.0 — 23 August 2026. Manuals current with this issue: 188 of 290.

§0 — READ THIS FIRST (HOW TO USE THIS MANUAL)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the engine law as seated (register contents line; the design, the concession, the provenance — verbatim) — §2 why it works (the physics, graded) — §3 the machine part by part — §4 the doctrine record — §5 the grade record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim, register no. 357): 'Phase 293: **Closed-Room Capillary-Wick Evaporation Engine and Bimetal-Band Unloader — 'the sneaky thermal engine'.** [NEW MACHINE — HIS INVENTION, SEATED 23 AUGUST 2026, SITTING 584 ON HIS WORD ('you can put that up as 293'); designed sitting 581, graded sittings 581-582, origin seated sitting 583] A sealed clear-polycarbonate room — concave walls and floor over a 2m sump; a 2-inch capillary wick tube lifts sump water to a splayed wick with dryer/water-sucker ring; evaporation does the work stroke; a 10-11m lever carries a 10-ton release onto a flywheel; the bimetal special band is the unloader — therefore the fuel line. Graded on the energy ledger: NOT a violator — an ambient-thermal engine tapped into the room's breathing gradient (isothermal room: stalls at equilibrium — affirmed by

the audit, conceded by the Author). Perpetual operation, never perpetual creation — the conflation thesis affirmed (sitting 582). Provenance: peltiers, two plates, a dentist's terrarium (sitting 583). Standing classification, his own concession: 'a sneaky thermal engine'. Rough ledger as graded: ~500 kJ/cycle at 1 cycle/year ≈ 0.14 kWh/yr — 'zero value other than to prove the point' (his words, affirmed).
[FULL PHASE SECTION + MANUAL GP-IM-293 SEATED 23 AUGUST 2026, SITTING 591, ON HIS ORDER]'

THE DESIGN (VERBATIM, SITTING 581)

'Ok we take a room we will say 500 square metres clear poly carbonte sealed but not a vacuum, in fact not airtight but it has a moisture lock, air can get in and out but not moisture. the roof is domed the walls concave or curving to the floor, the floor is concave draining to a 2 metre sump in the floor above the sump is a ten metre lever centre, at the sort end is a generator flywheel that is spun when the short end rises, a release switch releases the lever when the pressure from the tip hits 10 tons, so not a toy. hmmmm so how do you get 1 ton on the top end of the 11 metre lever energy free and get it to repeat??? repeat??? oh did i say that already? easy, the end of the lever hangs an aluminum ibc container empty, it has a speical band around the top six inches, the matching weight is on the short end of the lever of course no cheating. a 2 inch id clear tube in the sump goes to the top of the lever over into a small opening the ibc, no its not condensation but it is a sink condenser, its a 2 inch capillary wick with aluminum thread braided in for osmosis inside the tube. when the tank fills over the year or whatever it would take, lever solenoid mech hits 10 ton, 10 ton power falls 5metres, spins fly wheel, ibc hits ground, solenoid valve opens water releases valve closes when lever fully resets, water drains back to sump. repeat. the tube submersed is the key, osmosis fails if it can dry out, the water via osmosis or evaporation only has one way out, the trick is inside the container the wick is splayed out around against the inside the top circumference of the ibc along the aluminum wall at the special band point, in stripes around the tank as part of the tank are thin equal strips of bimetals with aluminum in-between, this band is vertical to the walls of the rest of the tank, bi metals heat and cool at different rates, but all sink to the wall forcing the water to let go of the primary wick, like the edges of frayed jeans, it becomes the path of least resistance, the moisture cannot escape, because the tube is sealed to the ibc, and the wick and tube prevent pressure lock. the 45 degree helps with easy osmosis. and a 2 metre sump accounts for room condensation etc. zero value other than to prove the point sometimes a closed system is why things work, our upper atmosphere is the ibc, our planet the sump, one uses osmosis, one uses convection it can be inside no sun, no solar energy required you're welcome Im here all week The Mighty Quinn'

THE CONCESSION (VERBATIM, SITTING 582)

'though it doesn't need clear walls it still works anyway because heat rises, you are correct so its still a sneaky thermal engine. the point is closed systems and perpetual motion are too often conflated'.

THE PROVENANCE (VERBATIM, SITTING 583)

'i could build dozens of closed systems machines, but they all have energy provided by nature, not batteries or electricity or engines. Do you want to know its origin? i was working with peltiers and sink devices for a while as a hobby and at electrotherm, i also built battery prototypes and capacitor at home alternating metal variation stacks, then learned how a thermocouple works bi metal tips. went home put an ally plate and copper plate together, put water about a ten cent piece on a copper plate an all plate of equal thickness and across the bimetal, they were seal with sink gel between them, water evaporated off the bimetal plates considerable faster, probably a well known thing, anyway, didnt think much of it until i was watching a terrarium in a dentists office one day just cycling the condensations, and thought what a really cool little engine. hence the dryer/water sucker ring for the splayed wick'

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE LIFT IS REAL; THE FREE LUNCH IS NOT. Capillarity can lift water ten metres — trees lift it a hundred — but surface tension holds what it lifts: water does not leave a wick for free. That is why the bimetal band is the heart of the machine and the audit's named cashier: bimetals work only on temperature CHANGE, so the band is the release mechanism, and therefore the fuel line. The band's alternating bimetal strips sink heat to the aluminium wall, changing the local wetting balance and forcing the water off the primary wick — 'like the edges of frayed jeans', the path of least resistance — with the dryer/water-sucker ring assisting the hand-off.

THE TWO-CASE VERDICT (the grade in one paragraph): (a) a truly isothermal room — the headspace saturates, the wick sits wet, the bimetals sit flat; the machine equilibrates and stalls: no violation, and no machine. (b) a room that breathes — day/night or seasonal swing, even indoors — the band flexes, unloads the wick, fills the IBC: a real, buildable, honest ambient-thermal engine, and bounded. 'Heat rises' is affirmed as one more open channel by which the gradient arrives — the machine has more taps than its builder first counted, and every one of them has a name.

THE CONFLATION THESIS, AFFIRMED OUTRIGHT (sitting 582): conservation bans perpetual ENERGY CREATION; it has never banned perpetual OPERATION. A river runs forever and breaks nothing, because its tap is the sun. Any machine tapped into an open gradient — solar, geothermal, tidal, or a room's thermal swing — may run perpetually and proudly. What no machine may do is run from nothing. THE TERM NOTE: 'osmosis' is used loosely in the design text — no semipermeable membrane or solute gradient appears; the mechanism is capillarity plus thermal-cycle unloading. His word stands verbatim per the law; the bench should say capillarity.

§3 — THE MACHINE, PART BY PART

- **The room:** ~500 m² clear polycarbonate, sealed but not a vacuum — not even airtight; moisture-locked: air in and out, moisture not. Domed roof; walls concave, curving to a concave floor draining to a 2-metre sump (the sump also takes room condensation).
- **The lever:** centre-pivoted, ten to eleven metres; generator flywheel on the short end; the matching weight on the short end — 'no cheating'. Release switch trips when tip pressure hits 10 tons — 'not a toy'.
- **The tank:** an empty aluminium IBC hanging on the long end; the special band around its top six inches — thin equal bimetal strips with aluminium between, vertical to the wall.
- **The lift line:** a 2-inch-ID clear tube from the sump, up over the lever, into a small opening in the IBC; inside, a 2-inch capillary wick with aluminium thread braided in; the tube stays submerged — 'osmosis fails if it can dry out'; the sealed tube-and-wick prevent pressure lock; the 45 degrees helps the climb.
- **The unload:** inside the tank the wick is splayed around the top inner circumference at the band point, in stripes; the bimetals heat and cool at different rates but all sink to the wall, forcing the water to let go of the primary wick; the dryer/water-sucker ring (born in the dentist's-terrarium moment) assists.
- **The cycle:** the wick lifts, the band unloads, the IBC fills over the year or whatever it takes; the solenoid mech hits 10 tons; 10 tons falls 5 metres and spins the flywheel; the IBC grounds; the solenoid valve opens, the water drains back to the sump; the valve closes when the lever fully resets. Repeat.

§4 — THE DOCTRINE RECORD (WHAT THIS MACHINE IS FOR)

THE STANDING EXHIBIT. This machine stands in the file as the standing exhibit of the distinction its builder named: not a perpetual-motion fraud, not a closed-system miracle — a perpetual-operation ambient engine with a named tap. THE QUINN-HAWKING LESSON, FINAL FORM AS GRADED (sitting 582): Newton's laws as guidelines — affirmed within their regime; conservation of energy — not a guideline, time-symmetry itself; perpetual motion — reclassified by his own thesis: perpetual creation is banned, perpetual operation is engineering. The dispute that opened at sitting 577 closed at 582, complete, both sides' words verbatim and the physics intact on both sides.

THE PLANET ANALOGY, HIS, SEATED: 'our upper atmosphere is the ibc, our planet the sump, one uses osmosis, one uses convection' — and the audit's sharpening: the ocean-to-cloud cycle is solar-driven, an external gradient with a named tap. The analogy concedes the machine's own verdict.

§5 — THE GRADE RECORD

NO VET BATCH REACHES HERE — the third-party vet batches run through phase 190. THE SITTINGS ARE THIS PHASE'S GRADE: designed sitting 581, graded sittings 581-582, origin seated sitting 583, taken into the file sitting 584 on his word ('you can put that up as 293'). The grade: an ambient-thermal engine — real, buildable, honest, bounded. THE ROUGH LEDGER, AS GRADED: 10 tons x 5 m = ~500 kJ per cycle; at his one-cycle-per-year, ~0.14 kWh/year — 'zero value other than to prove the point' (his words, affirmed correct). THE BUILD MECHANICS, affirmed without reservation: the lever arithmetic, the flywheel release, the drain-and-reset, the submerged-wick discipline, the condensation sump. THE BUILDABLE VERSION, offered by the audit and seated: open the loop and this exact hardware is a legitimate atmospheric-water gravity-battery demonstrator — an engine, graded green as an engine.

§6 — BUILD SEQUENCE

- 1. Build the room: polycarbonate panels, domed roof, concave walls and floor; fit the moisture lock (air passes, vapour does not); drain the floor to the 2 m sump.
- 2. Set the pivot and hang the lever; fit the flywheel and generator on the short end; fit the matching weight — no cheating.
- 3. Fit the release: solenoid mech and switch, tripped at 10 tons tip pressure; the drain valve on the IBC, closing when the lever fully resets.
- 4. Hang the IBC on the long end; fit the special band — bimetal strips, aluminium between, vertical, around the top six inches; fit the dryer/water-sucker ring.
- 5. Run the lift line: 2-inch clear tube, sump to IBC; braid the aluminium thread into the wick; splay the wick at the band point inside the tank; keep the tube submerged — it must never dry.
- 6. Charge the sump, seal the room, and wait for the weather to do the work. Instrument the first cycles: the bench numbers are owed (§9).

§7 — CROSS-PHASE

No ship system depends on this machine — it is a demonstrator, not a fitment; nothing else in the file draws on it and it draws on nothing but the weather. Its doctrine — perpetual operation from a named open gradient — is the file's standing answer wherever 'perpetual motion' is spoken. The open-loop variant (the audit's offer, seated): the same hardware as an atmospheric-water gravity battery belongs earth-side, 'for the people on earth', alongside Frost Free Fred (unphased, his word if it joins).

§8 — DO-NOT-BUILD

- Do NOT build it as, or sell it as, a perpetual-creation machine — the file's own grade kills that reading: seal the room truly isothermal and it equilibrates and stalls. The moisture lock seals water, not energy.
- Do NOT let the lift tube dry out — the lift fails the moment the wick can dry; submerged is the discipline.
- Do NOT skip the matching weight or the 10-ton release discipline — 'not a toy' is the spec, not a boast; ten tons falling five metres is a workplace hazard with a flywheel attached.
- Do NOT write 'osmosis' on the bench sheets — the mechanism is capillarity plus thermal-cycle unloading (the term note, §2); his word stands verbatim in the law, the bench uses the true one.

§9 — OWED

OWED (bench, all his to run or delegate): the wick lift rate against the room's thermal swing; the band's unload rate per degree of change; the true cycle period in a real breathing room (his estimate: a year); measured energy per cycle against the ~500 kJ arithmetic; the dryer-ring's contribution, with and without; and the open-loop atmospheric-water demonstrator variant, if he wants it specced. No vet batch will come — the sittings graded it; the bench confirms it.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: MANUAL GP-IM-293, v1.0 — CURRENT — an ambient-thermal engine, graded in sittings 581-582 (the sittings ARE the grade; no vet batch reaches here); seated law, master v2.1 (numerical print order). Manuals current with this issue: 188 of 290. Built 23 August 2026 on the author's order of the same day (sitting 591). First version until publication (first-version law: until he publishes it, it is still first version).

PROVENANCE — THE ORDER, VERBATIM (seated in sitting 591): 'put up 293 now and i will give you 294'. The phase's own chain, all seated in the master: the design, verbatim (sitting 581 — 'Ok we take a room we will say 500 square metres...'); the grade on the energy ledger (sitting 581) against the boundary fixed in sitting 580; the concession and the conflation thesis (sitting 582 — 'so its still a sneaky thermal engine'); the origin (sitting 583 — peltiers, two plates, a dentist's terrarium); the taking into the file on his word (sitting 584 — 'you can put that up as 293'); the full phase section and this manual (sitting 591). The manual program's standing orders chain (sittings 519, 537, 557, 561, 562, 564, 566, 572, 576, 587, 590) stands as seated in the manuals before this one.

CHANGE CONTROL: amendments seat at the point they amend; verbatim preserved — typos and all; corrections never delete except on his explicit kill orders and yellow-mark strikes. No history lessons in a

workshop manual — the builders get the current machine; the full change record lives in the master and the restart (his ruling, 22 August 2026: 'i deleted as you see, again history lessons are not relevant to the builders'). Next update triggers: the bench numbers, or phase 294's arrival, or his word.

END OF MANUAL — PHASE 293